

Frontend and Backend codes

Notes

> Backend = PHP Logic
> Frontend = HTML, CSS, a bit of javascript
> always use .php

Login page scenario

FRONTEND

```php
<html>
    <head>
        <title>Login Module</title>
    </head>
    <body>
        <form action="login.php" method="post">
            <table>
                <tr>
                    <td>Enter Username:</td>
                    <td><input type="text" name="username" required></td>
                </tr>
                <tr>
                    <td>Enter Password:</td>
                    <td><input type="password" name="password" required></td>
                </tr>
                <tr>
                    <td><input type="submit" value="Login"></td>
                    <td><input type="reset" value="Clear"></td>
                </tr>
            </table>
        </form>
    </body>
</html>
```

> file name: home.php

BACKTEND

```php
<?php
    $username = $_POST['username'];
    $password = $_POST['password'];
    $dafaultUsername = "admin";
    $defaultPassword = "admin123";
    if($username == $dafaultUsername && $password == $defaultPassword){
?>
        <script>
            alert("Login Successful");
            window.location.href = "index.php";
        </script>
<?php
    }
    else{
?>
        <script>
            alert("Login Failed");
            window.location.href = "home.php";
        </script>
<?php
    }
?>
```

> file name: login.php

Status highlighting of record scenario

FRONTEND AND BACKEND

```php
<html>
    <head>
        <title>Record Management</title>
    </head>
    <body>
        <table>
            <thead>
                <tr>
                    <th>Course</th>
                    <th>Section</th>
                    <th>Status</th>
                </tr>
            </thead>
            <tbody>
                <tr
                    <?php
                        $stat = "1";
                        if($stat == 1){
                            echo "style='background: green;'";
                        }
                        else{
                            echo "style='background: red;'";
                        }
                    ?>
                >
                    <td>BSIT</td>
                    <td>1A</td>
                    <td>Active</td>
                </tr>
                <tr
                    <?php
                        $stat = "0";
                        if($stat == 1){
                            echo "style='background: green;'";
                        }
                        else{
                            echo "style='background: red;'";
                        }
                    ?>
                >
                    <td>BSCS</td>
                    <td>2B</td>
                    <td>Inactive</td>
                </tr>
                <tr
                    <?php
                        $stat = "1";
                        if($stat == 1){
                            echo "style='background: green;'";
                        }
                        else{
                            echo "style='background: red;'";
                        }
                    ?>
                >
                    <td>BSIS</td>
                    <td>3C</td>
                    <td>Active</td>
                </tr>
            </tbody>
        </table>
    </body>
</html>
```

> file name: record.php
> note: all codes between <?php ?> is backend

Registration module scenario (with javascript)

FRONTEND

```html
<html>
    <head>
        <title>Sample Registration</title>
        <link rel="stylesheet" href="style.css">
        <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
    </head>
    <body>
        <h2>Create Account</h2>
        <table>
            <tr>
                <td>Full Name:</td>
                <td><input type="text" name="fullname" id="fullname" placeholder="John Doe"></td>
            </tr>
            <tr>
                <td> </td>
                <td colspan=2><div id="fullname_error"></div></td>
            </tr>
            <tr>
                <td>Username:</td>
                <td><input type="text" name="username" id="username" placeholder="johndoe_123"></td>
            </tr>
            <tr>
                <td> </td>
                <td colspan=2><div id="username_error"></div></td>
            </tr>
            <tr>
                <td>Email:</td>
                <td><input type="email" name="email" id="email" placeholder="sample@email.com"></td>
            </tr>
            <tr>
                <td> </td>
                <td colspan=2><div id="email_error"></div></td>
            </tr>
            <tr>
                <td>Phone:</td>
                <td><input type="tel" name="phone" id="phone" placeholder="09123456789"></td>
            </tr>
            <tr>
                <td> </td>
                <td colspan=2><div id="phone_error"></div></td>
            </tr>
            <tr>
                <td>Password (Min. 8 chars, 1 letter & 1 number):</td>
                <td><input type="password" name="password" id="password" placeholder="Min. 8 chars, 1 letter & 1 number"></td>
            </tr>
            <tr>
                <td> </td>
                <td colspan=2><div id="password_error"></div></td>
            </tr>
            <tr>
                <td>Confirm Password:</td>
                <td><input type="password" name="confirm_password" id="confirm_password" placeholder="Re-type password"></td>
            </tr>
            <tr>
                <td> </td>
                <td colspan=2><div id="confirm_password_error"></div></td>
            </tr>
            <tr>
                <td colspan="3">
                    <button class="btn" id="reg_btn">Register Now</button>
                </td>
            </tr>
        </table>
        <script src="validation.js"></script>
    </body>
</html>
```

> file name: index.php

```javascript
$(document).ready(function() {

    $("#reg_btn").on("click", function(e) {
        let isValid = true;

        let fullname = $("#fullname").val();
        let username = $("#username").val();
        let email = $("#email").val();
        let phone = $("#phone").val();
        let password = $("#password").val();
        let confirm_password = $("#confirm_password").val();

        // Full Name
        if (fullname.length < 3) {
            $('#fullname_error').text("Full name must be at least 3 characters long.");
            isValid = false;
        } else if (!/^[a-zA-Z\s]+$/.test(fullname)) {
            $('#fullname_error').text("Full name can only contain letters and spaces.");
            isValid = false;
        }

        // Username
        if (!/^[a-zA-Z0-9_]{4,20}$/.test(username)) {
            $('#username_error').text("Username must be 4-20 characters long and can only contain letters, numbers, and underscores.");
            isValid = false;
        }

        // Email
        if (!/^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email)) {
            $('#email_error').text("Please enter a valid email address.");
            isValid = false;
        }

        // Phone
        if (!/^[0-9]{10,15}$/.test(phone)) {
            $('#phone_error').text("Phone number must be between 10 to 15 digits.");
            isValid = false;
        }

        // Password
        if (password.length < 8) {
            $('#password_error').text("Password must be at least 8 characters long.");
            isValid = false;
        } else if (!/(?=.*[A-Za-z])(?=.*\d)/.test(password)) {
            $('#password_error').text("Password must contain at least one letter and one number.");
            isValid = false;
        }

        // Confirm Password
        if (confirm_password !== password) {
            $('#confirm_password_error').text("Passwords do not match.");
            isValid = false;
        }

        if (isValid) {

            let url = "process.php?fullname=" + fullname +
            "&username=" + username +
            "&email=" + email +
            "&phone=" + phone +
            "&password=" + password +
            "&confirm_password=" + confirm_password;
            window.location.href = url;

        }
    });
});
```

> file name: validation.js

BACKEND

```php
<?php
    $fullname = $_GET['fullname'];
    $username = $_GET['username'];
    $email = $_GET['email'];
    $phone = $_GET['phone'];
    $password = $_GET['password'];
    $confirm_password = $_GET['confirm_password'];

    echo "Full Name: $fullname<br>";
    echo "Username: $username<br>";
    echo "Email: $email<br>";
    echo "Phone: $phone<br>";
    echo "Password: $password<br>";
    //Database insertion logic here
?>

```

> file name: process.php

Student Grade Evaluation System scenario (if-elseif-else)

FRONTEND

```php
<!DOCTYPE html>
<html>
<head>
    <title>Student Grade Evaluation</title>
</head>
<body>

    <h2>Student Grade Evaluation System</h2>

    <form action="scenario1result.php" method="post">

        <label>Student Name:</label>
        <input type="text" name="student_name" required>
        <br><br>

        <label>Subject:</label>
        <input type="text" name="subject" required>
        <br><br>

        <label>Score:</label>
        <input type="number" name="score" min="-100" max="200" required>
        <br><br>

        <input type="submit" value="Evaluate Grade">

    </form>

</body>
</html>
```

> file name: scenario1index.php

BACKEND

```php
<?php

$student_name = $_POST['student_name'];
$subject = $_POST['subject'];
$score = $_POST['score'];

if ($score > 100 || $score < 0) {
    $grade = "N/A";
    $remarks = "Invalid Score";
}
elseif ($score >= 90) {
    $grade = "A";
    $remarks = "Excellent - With Honors";
}
elseif ($score >= 80) {
    $grade = "B";
    $remarks = "Very Good";
}
elseif ($score >= 70) {
    $grade = "C";
    $remarks = "Good";
}
elseif ($score >= 60) {
    $grade = "D";
    $remarks = "Passed - Needs Improvement";
}
else {
    $grade = "F";
    $remarks = "Failed";
}

?>
```

```php
<!DOCTYPE html>
<html>
<head>
    <title>Grade Result</title>
</head>
<body>

    <h2>Student Grade Result</h2>

    <p><strong>Student Name:</strong> <?php echo htmlspecialchars($student_name); ?></p>
    <p><strong>Subject:</strong> <?php echo htmlspecialchars($subject); ?></p>
    <p><strong>Score:</strong> <?php echo $score; ?></p>
    <p><strong>Grade:</strong> <?php echo $grade; ?></p>
    <p><strong>Remarks:</strong> <?php echo $remarks; ?></p>

    <?php
    if ($grade == "A" && $score == 100) {
        echo "<p><strong>Perfect Score!</strong></p>";
    }
    ?>

    <br>
    <a href="scenario1index.php">Evaluate Another Student</a>

</body>
</html>
```

> file name: scenario1result.php

Online Shop Discount Calculator scenario (Nested if)

FRONTEND

```html
<!DOCTYPE html>
<html>
<head>
    <title>Online Shop Discount Calculator</title>
</head>
<body>

    <h2>Online Shop Discount Calculator</h2>

    <form action="scenario2result.php" method="post">

        <label>Customer Name:</label>
        <input type="text" name="customer_name" required>
        <br><br>

        <label>Total Purchase:</label>
        <input type="number" name="purchase" min="0" step="0.01" required>
        <br><br>

        <label>Member:</label>
        <input type="radio" name="member" value="Yes" required> Yes
        <input type="radio" name="member" value="No"> No
        <br><br>

        <input type="submit" value="Calculate Discount">

    </form>

</body>
</html>
```

> file name: scenario2index.php

BACKEND

```php
<?php

$customer_name = $_POST['customer_name'];
$purchase = $_POST['purchase'];
$member = $_POST['member'];

if ($member == "Yes") {

    if ($purchase >= 5000) {
        $discount = 20;
    } else {
        $discount = 10;
    }

} else {

    if ($purchase >= 5000) {
        $discount = 5;
    } else {
        $discount = 0;
    }
}

$discount_amount = $purchase * ($discount / 100);
$final_amount = $purchase - $discount_amount;

?>
```

```php
<!DOCTYPE html>
<html>
<head>
    <title>Discount Result</title>
</head>
<body>

    <h2>Discount Calculation Result</h2>

    <p><strong>Customer Name:</strong>
        <?php echo htmlspecialchars($customer_name); ?>
    </p>
    <p><strong>Total Purchase:</strong>
        ₱<?php echo number_format($purchase, 2); ?>
    </p>
    <p><strong>Member:</strong>
        <?php echo $member; ?>
    </p>
    <p><strong>Discount:</strong>
        <?php echo $discount; ?>%
    </p>
    <p><strong>Discount Amount:</strong>
        ₱<?php echo number_format($discount_amount, 2); ?>
    </p>
    <p><strong>Final Amount:</strong>
        ₱<?php echo number_format($final_amount, 2); ?>
    </p>

    <br>
    <a href="scenario2index.php">Calculate Another Purchase</a>

</body>
</html>
```

> file name: scenario2result.php

Food Menu Price List scenario (switch case)

FRONTEND

```php
<!DOCTYPE html>
<html>
<head>
    <title>Food Menu Price List</title>
</head>
<body>

    <h2>Food Menu</h2>

    <form action="scenario3result.php" method="post">

        <label for="code">Select Menu Code:</label>
        <select name="code" id="code">
            <option value="A">A - Chicken Adobo + Rice</option>
            <option value="B">B - Pork Sinigang + Rice</option>
            <option value="C">C - Beef Steak + Rice</option>
            <option value="D">D - Vegetable Curry + Rice</option>
        </select>

        <br><br>

        <label for="qty">Quantity:</label>
        <input type="number" name="qty" id="qty" min="1" required>

        <br><br>

        <button type="submit">Calculate Total</button>

    </form>

</body>
</html>
```

> file name: scenario3index.php

BACKEND

```php
<?php
$code = $_POST['code'];
$qty = $_POST['qty'];

switch ($code) {
    case "A":
        $meal = "Chicken Adobo + Rice";
        $price = 75;
        break;

    case "B":
        $meal = "Pork Sinigang + Rice";
        $price = 85;
        break;

    case "C":
        $meal = "Beef Steak + Rice";
        $price = 95;
        break;

    case "D":
        $meal = "Vegetable Curry + Rice";
        $price = 60;
        break;

    default:
        echo "<h2>Invalid Menu Code.</h2>";
        exit;
}
if ($qty <= 0) {
    echo "<h2>Error: Quantity must be greater than 0.</h2>";
    exit;
}

$total = $price * $qty;
?>
```

```php
<!DOCTYPE html>
<html>
<head>
    <title>Order Result</title>
</head>
<body>

    <h2>Order Summary</h2>

    <p><strong>Menu Code:</strong> <?php echo $code; ?></p>
    <p><strong>Meal:</strong> <?php echo $meal; ?></p>
    <p><strong>Price:</strong> ₱<?php echo $price; ?></p>
    <p><strong>Quantity:</strong> <?php echo $qty; ?></p>
    <p><strong>Total:</strong> ₱<?php echo $total; ?></p>

    <br>

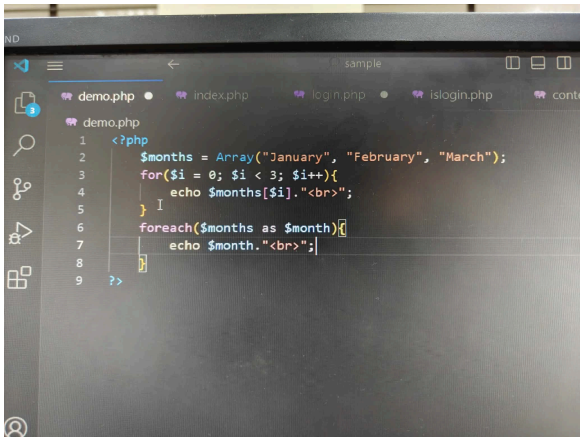
    <a href="scenario3index.php">Back to Menu</a>

</body>
</html>
```

> file name: scenario3result.php

USE OF ARRAY scenario

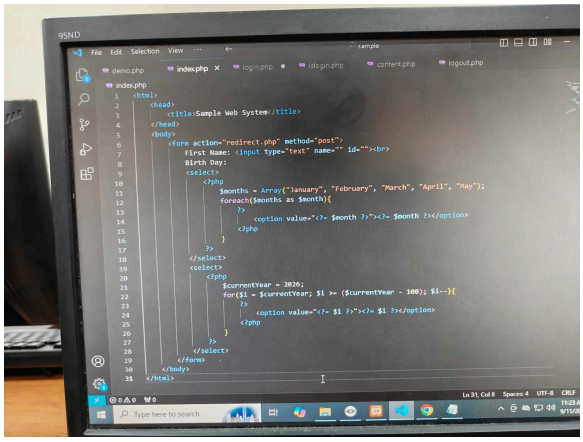
BACKEND



> file name: demo.php

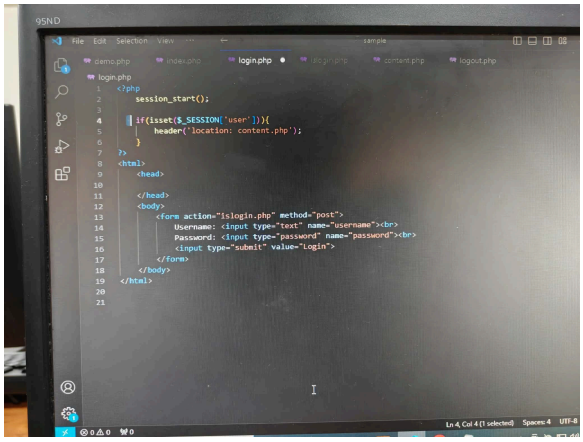
Sample web system scenario

FRONTEND

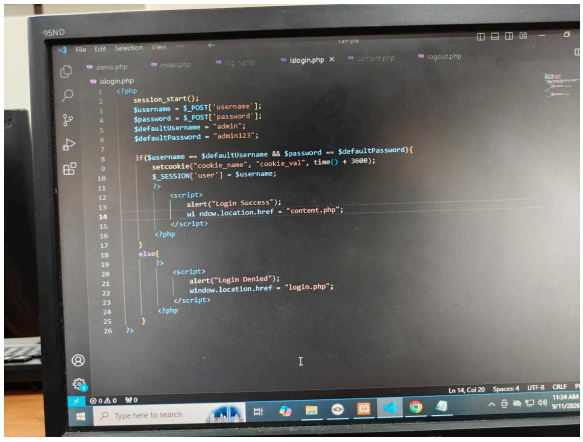


> file name: index.php

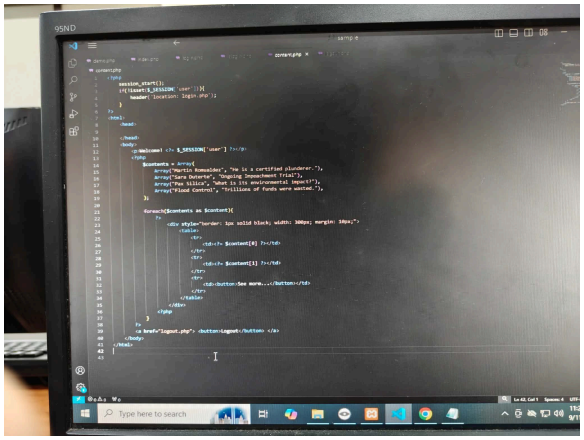
BACKEND



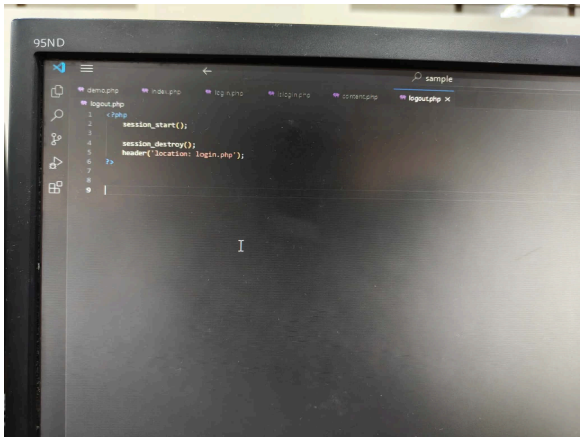
> file name: login.php



> file name: islogin.php



> file name: content.php



> file name: logout.php